

Abstract

The understanding of the movement of water through soil has many applications. The Richards Equation has become one of the most common ways to describe and quantify the transient movement of water in unsaturated porous media which uses the unsaturated hydraulic conductivity. The unsaturated hydraulic conductivity is important for both physical and statistical models. Hydraulic conductivity is determined from a water retention curve which is a plot of soil moisture verses unsaturated hydraulic head. The water retention curve, however, is often difficult to obtain in the laboratory and difficult to apply to soil moisture data in the field.

A fixture called The Travel Assembly has been developed to hold a soil sensor in a soil sample during the laboratory development of a soil water retention curve. The Travel Assembly is a fixture that holds the soil sample with a HydraProbe, so that it can be pressurized up to 15 bar. The resulting soil water retention curve can then be correlated with the HydraProbe. Upon completion of the soil water retention curve, HydraProbes in the field in the same soil tested are able to measure soil moisture very accurately with a custom calibration, and unsaturated hydraulic head. With this information and the unsaturated hydraulic conductivity can then be calculated using the van Genuchten equation.

Six soil samples were placed in the Travel Assembly and evaluated at saturation and 10 pressures ranging from 0.01 to 5 Bar. The technique was perfected to generate a soil water retention curve with minimal error.

Water Retention Curve and the van Genuchten Equation.

Soil water movement can be explained and modeled using the plot of soil moisture vs. hydraulic head.

The van Genuchten Equation is the mathematical relationship between the hydraulic head and soil moisture and provides a numerical solution for unsaturated hydraulic conductivity.

$$\theta = \frac{\theta_s - \theta_r}{[1 + |ah|^n]^m} + \theta_r \quad [1]$$

$$h = \frac{\left(\left[\frac{\theta_s - \theta_r}{\theta - \theta_r} \right]^{1/m} - 1 \right)^{1/n}}{\alpha} \quad [2]$$

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad [3]$$

$$K = K_s S_e^l \left[1 - \left(1 - S_e^{1/m} \right)^m \right]^2 \quad [4]$$

Monitoring the Progress of the De-Watering inside the Extractor

Without a sensor in the sample, equilibrium can be difficult to determine. If the experiment is terminated too soon, error can be introduced and it would take extra labor to re-run the experiment. The soil moisture can be monitored while it is dewatering to inform the user of the progress.

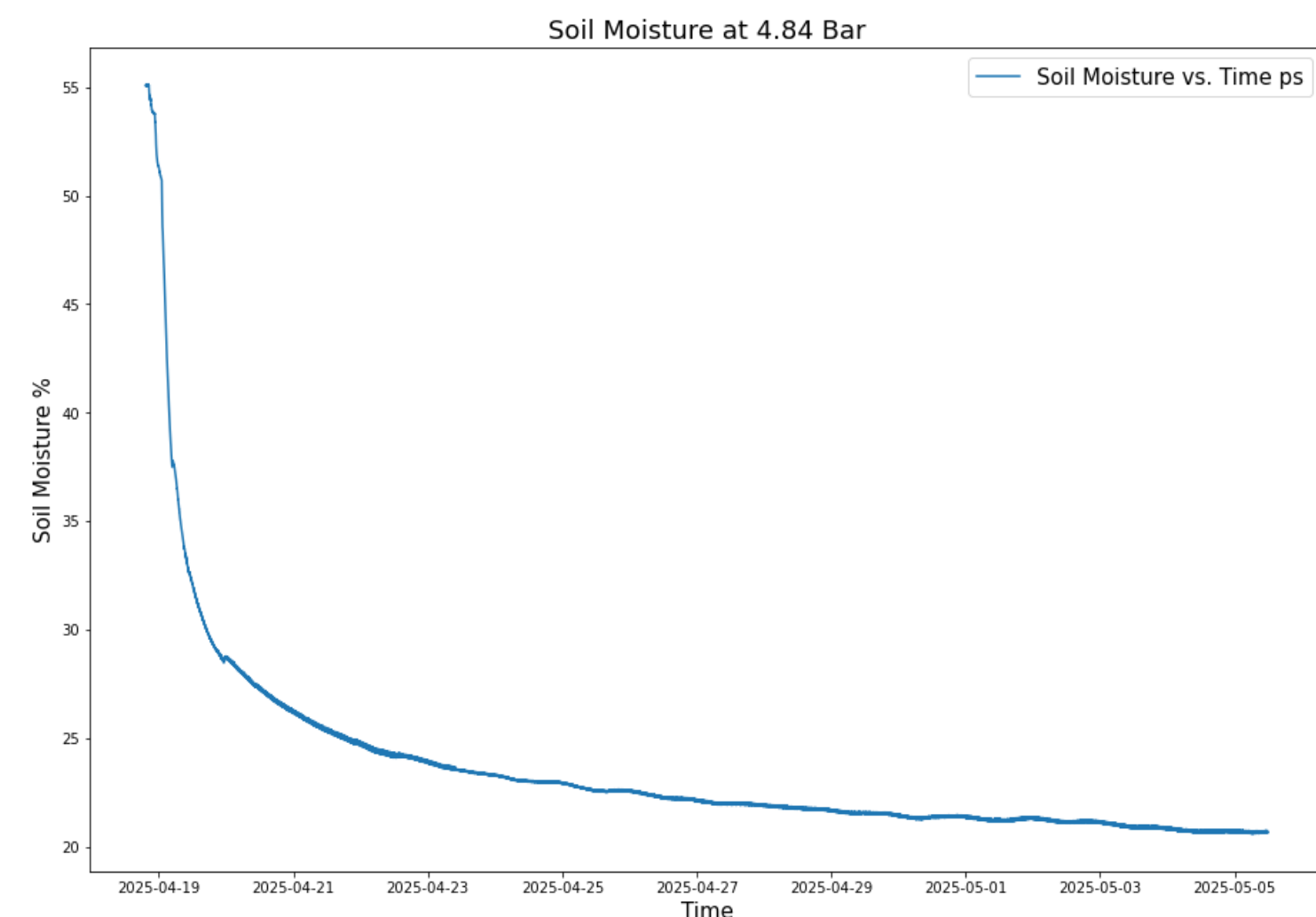


Figure [3], Soil is saturated on April 18th and brought to 5000 kPa. The Sample reaches equilibrium on May 5th, 18 Days.

Determination of Equilibrium and Completion Time

Some soil samples at higher pressures take many day or weeks to reach equilibrium. A regression can be made to estimate the end time and estimate the soil moisture at equilibrium to cut days or weeks off of the testing time.

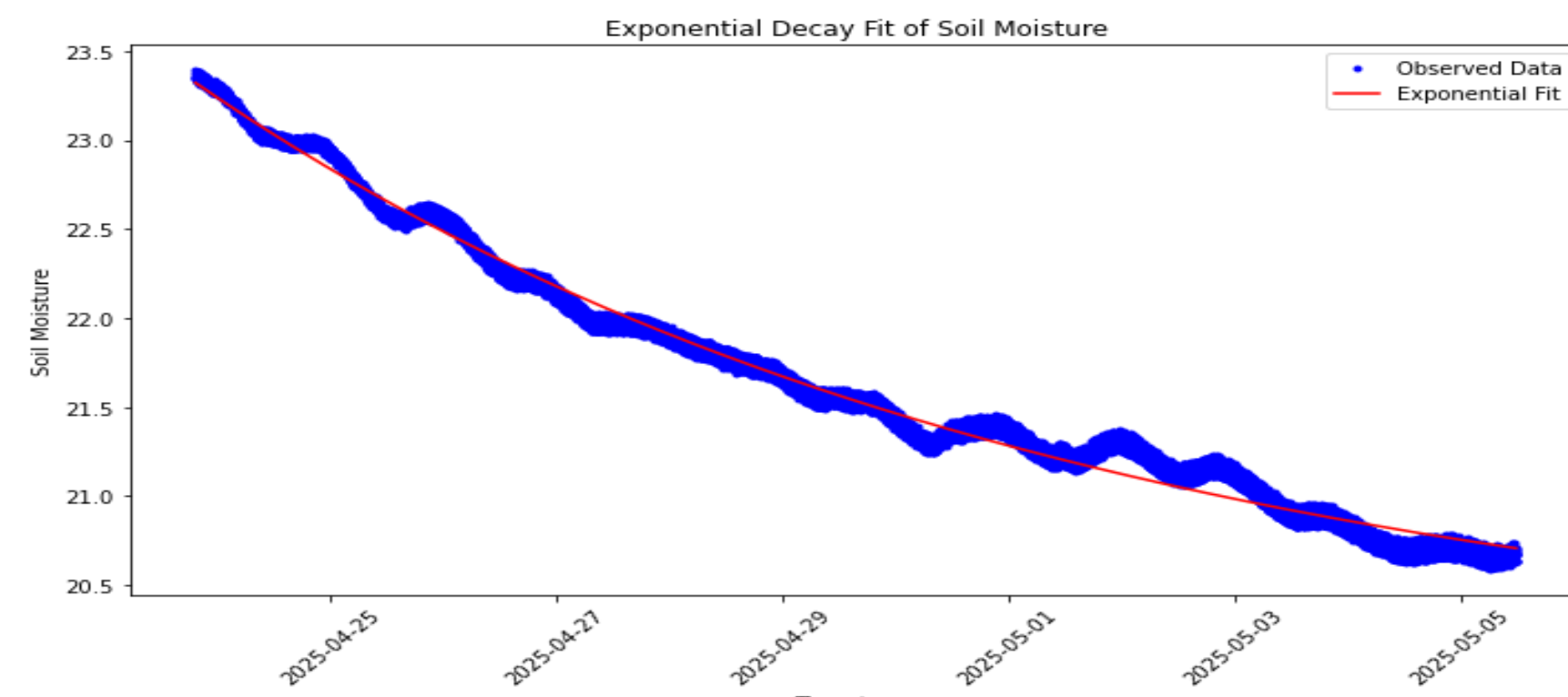


Figure [4], Exponential regression of de-watering to estimate the completion

Exponential Decay of Soil Moisture

$$\theta = Ae^{-kt} + C \quad [5]$$

A = 3.3279
k = 0.0000002
C = 19.9941

The R² between soil moisture and the curve fit decay of soil moisture every 6 hours is 0.992

Custom Calibration of HydraProbe Soil Sensor

A custom calibration of the HydraProbe soil sensor can be performed from the data collected in the extractors and low tension tables.

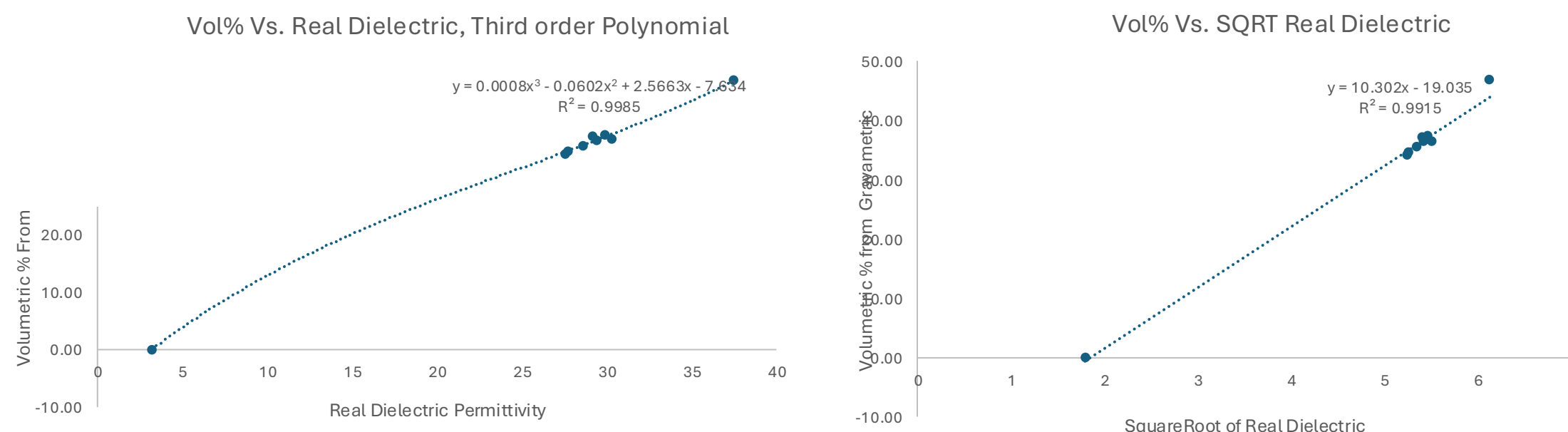


Figure [5 a&b], Custom calibration with 3rd order polynomial and semi-linear regressions

Custom Calibration N=9, Three Trials Each			
Method	Formula	R ²	RMSE
Factory (Seyfried Equation)	$\theta = A + B\epsilon_r^{0.5}$	0.9863	+/- 1.44%
Third Order Polynomial	$\theta = A + B\epsilon_r + C\epsilon_r^2 + D\epsilon_r^3$	0.9985	+/- 0.16%
Semi-Linear, Square root of real Dielectric	$\theta = A + B\epsilon_r^{0.5}$	0.9915	+/- 0.38%

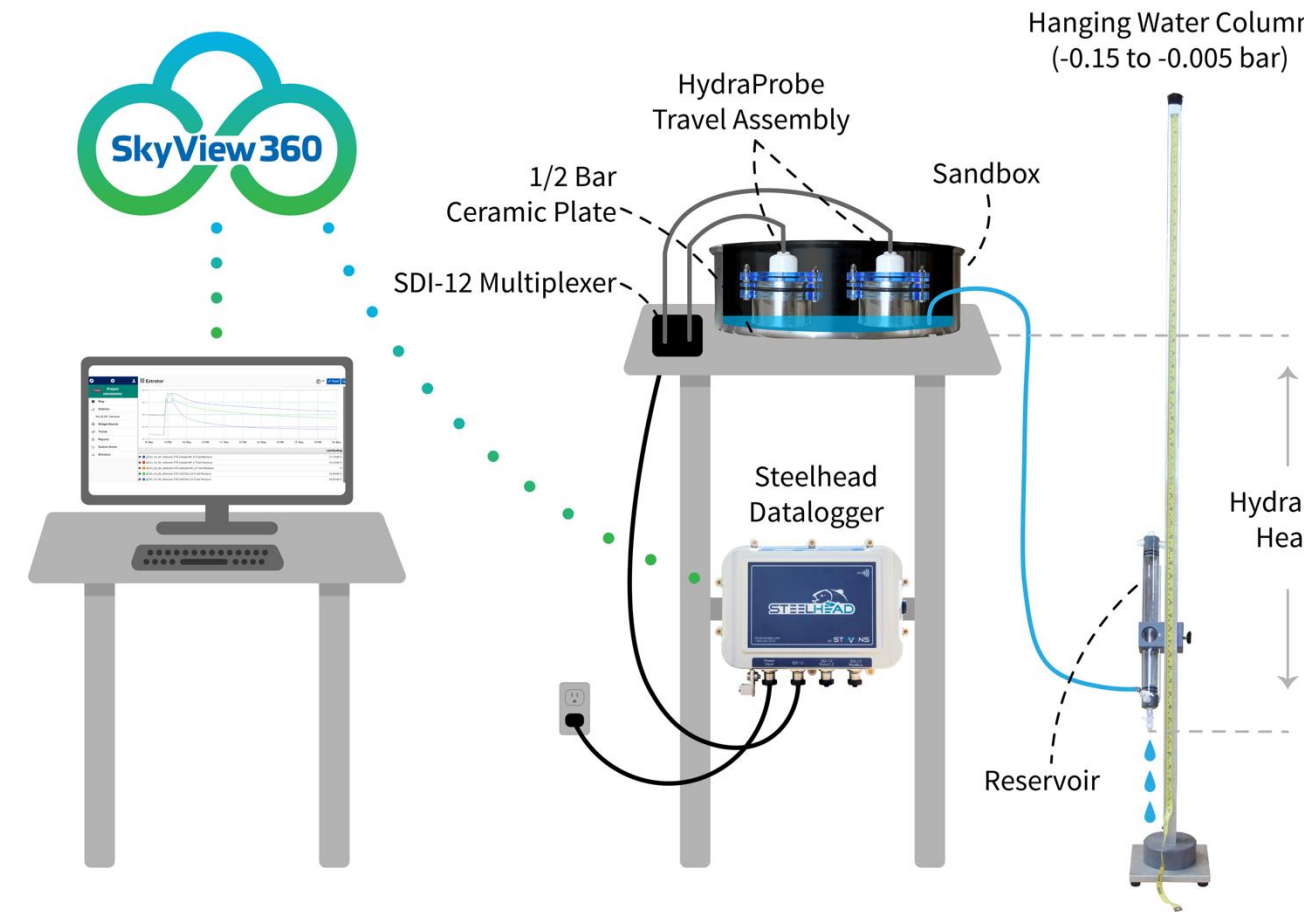


Figure [6], Diagram of low-tension lab setup complete with Stevens Steelhead and HydraProbe Travel Assembly.



Figure [8], Low -tension lab setup

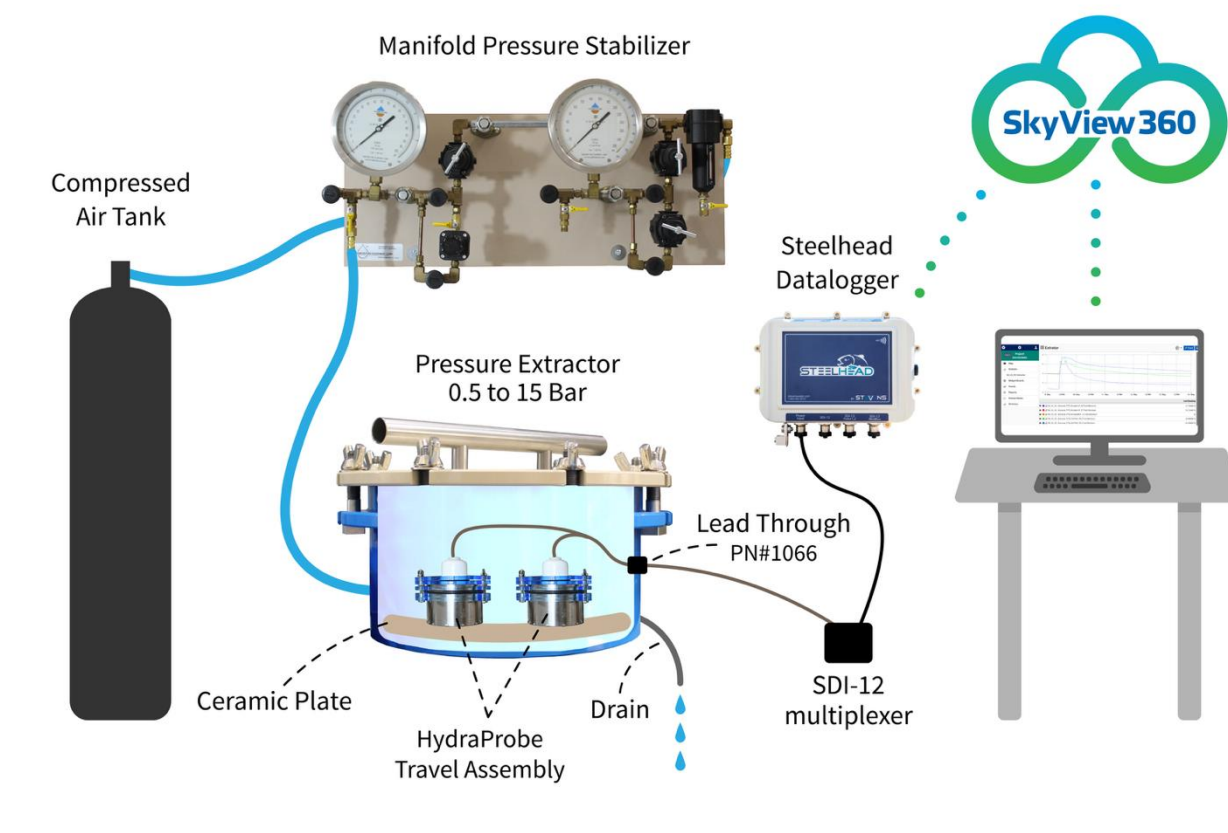


Figure [7], Diagram of pressurized lab setup complete with Stevens Steelhead and HydraProbe Travel Assembly



Figure [9], Hanging water column up to 5 meters high with reservoir.

Soil Water Retention Curve Soil Moisture Vs. Hydraulic Head

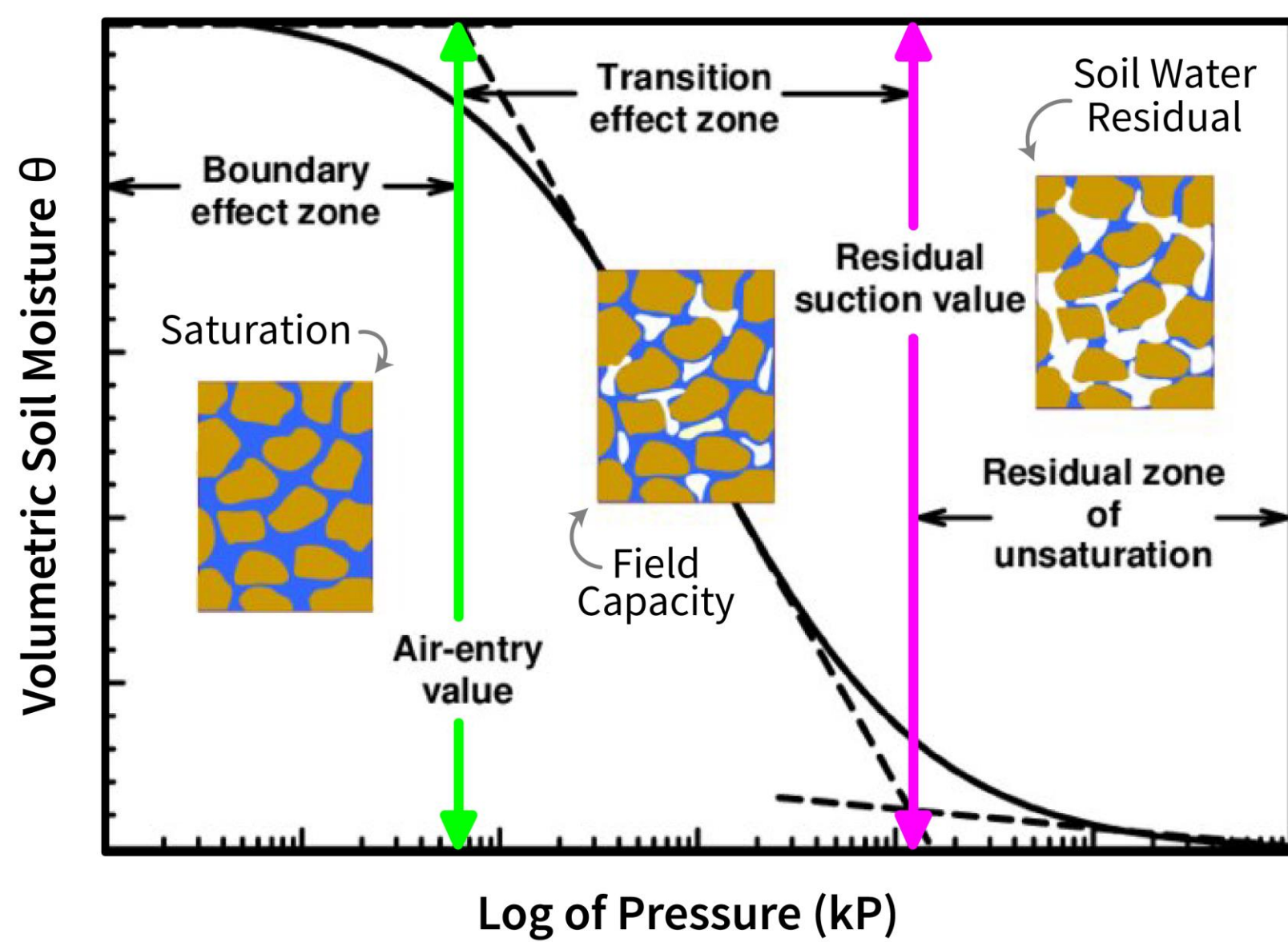


Figure [1], Features of a Soil water retention curve.

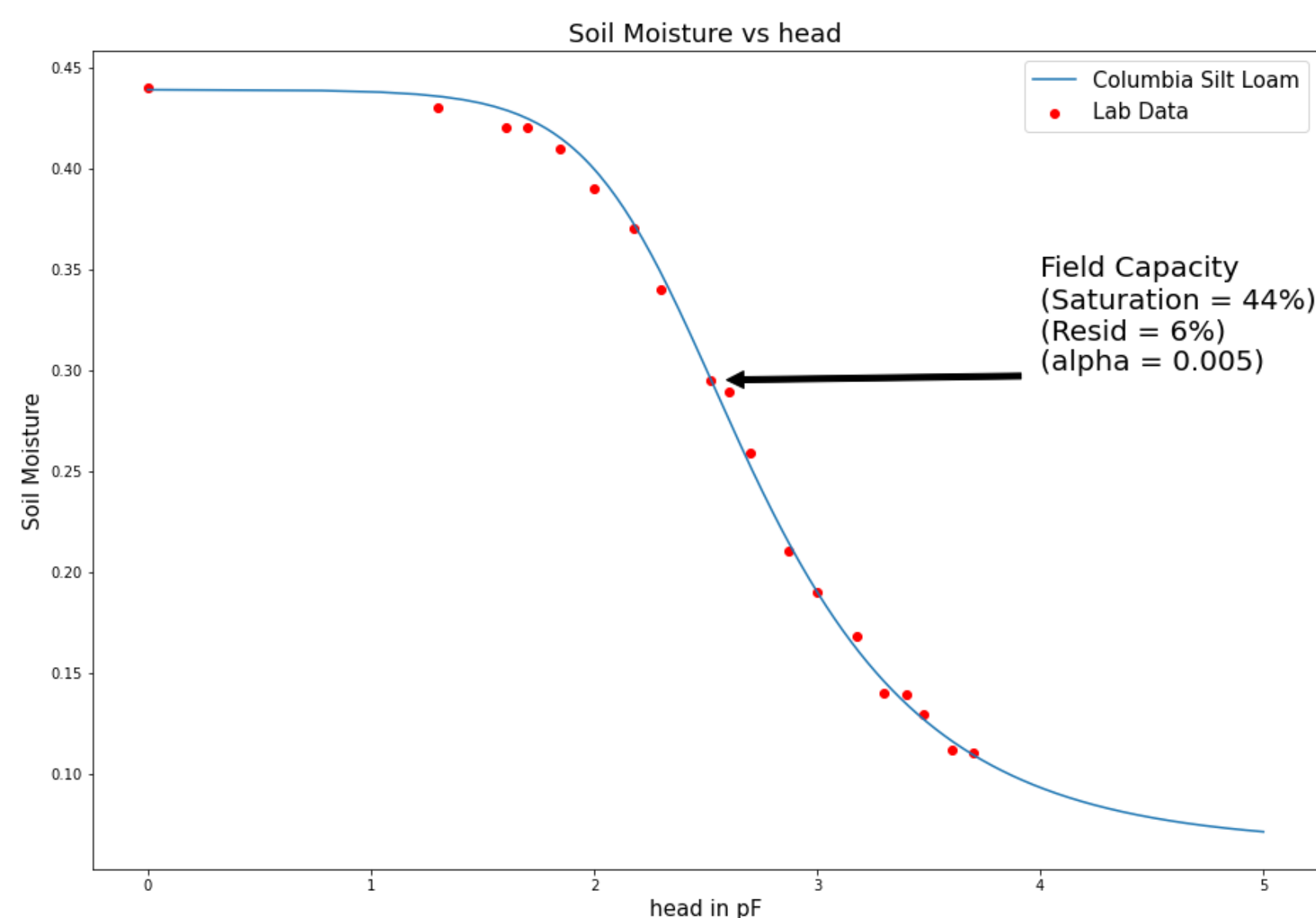


Figure [2], Soil water retention curve of Columbia Silt Loam

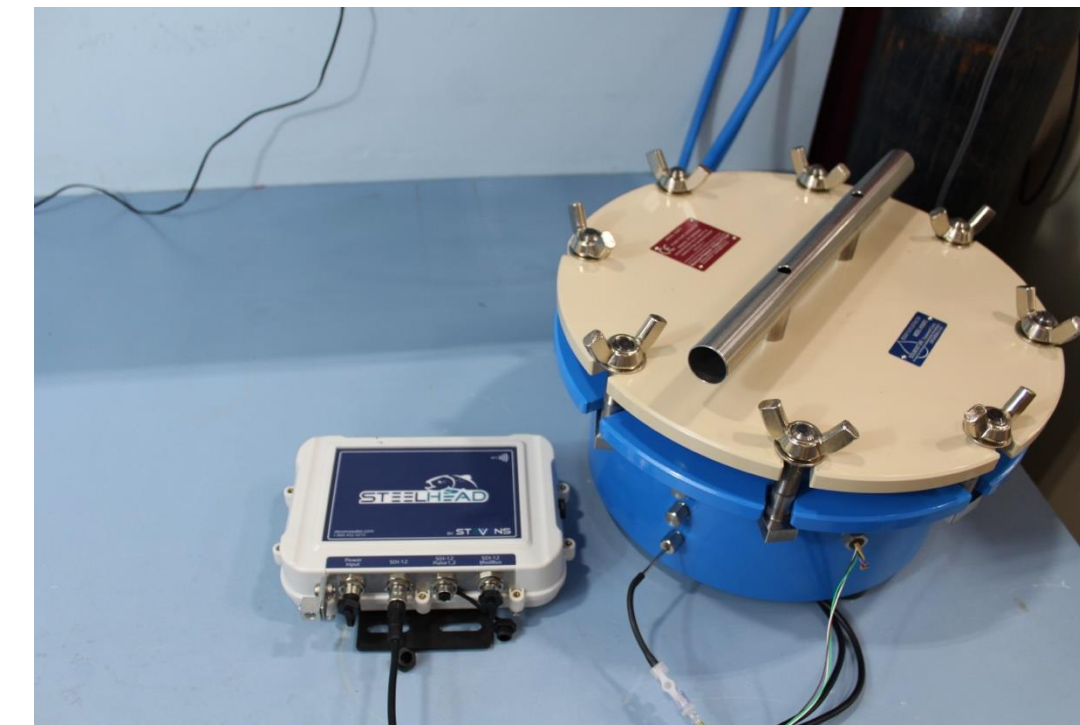


Figure [10], High Pressure Extractor that goes from 500 to 15000 kPa. Internal sensors can be connected to a Steelhead Logger



Figure [11], HydraProbe Travel Assembly in a Tempe Cell for the determination of saturation.



Figure [12], Travel Assembly. Metal ring holding soil sample and HydraProbe. The bottom has a mesh which allows water to pass in and out of the soil sample.

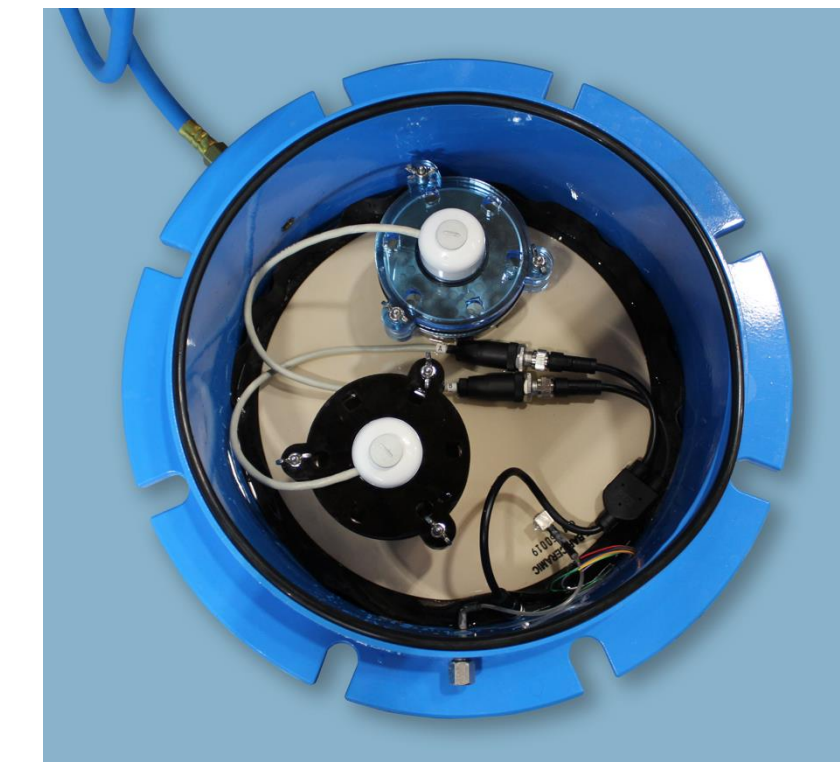


Figure [13], Travel Assembly in a pressurized extractor sitting on a ceramic plate.